JUST Undergraduate Final Admission System

A Project Report Submitted

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Preface

On the eve of the beginning of civilization, human beings had been striving tirelessly to discover, explore, and innovate unknown aspects of the world. This inherent curiosity and desire for advancement have driven humanity toward continuous progress and development. Over time, human beings have not only acquired theoretical knowledge but have also emphasized the importance of practical implementation, which plays a crucial role in solving real-world problems and improving the quality of life.

In the modern era, the field of Computer Science and Engineering has emerged as one of the most dynamic and influential disciplines, contributing significantly to technological advancement and digital transformation. The philosophy of the Computer Science and Engineering department is centered on equipping students with both theoretical understanding and practical skills. To achieve this goal, the department has introduced thesis and project programs that encourage students to engage in real-life problem-solving and innovative system development.

As a part of fulfilling the academic requirements of the undergraduate program, I have undertaken a project titled **"JUST Undergraduate Final Admission System"**. This project aims to develop a comprehensive web-based platform that simplifies and streamlines the admission process for undergraduate students. It focuses on efficiency, transparency, and user-friendliness, ensuring a better experience for both applicants and administrators.

Throughout the development of this project, I have conducted extensive research, analysis, and observation to understand the existing admission processes and identify their limitations. This has enabled me to design and implement a system that addresses real-world challenges effectively. The project has provided me with valuable opportunities to enhance my technical skills, problem-solving abilities, and understanding of software development methodologies.

I believe that this project will not only serve as an academic accomplishment but also contribute positively to the automation and digitalization of admission systems. The knowledge and experience gained during this work will undoubtedly play a vital role in my future career and professional development.

Contents

Chapter 1

Introduction

1.1 Background

Admission of a university is an indispensable activity in every educational activity and is as old as education itself. With more students applying for admission than in previous years, admission process for admissions officers has become increasingly time-consuming. More applications resulted in heavier paperwork and processing challenges. Every admission is routed through various faculty for evaluation, and the manual admission process caused difficulty in admission, and archiving.

In order to speed up and simplify the admission process, the development of the Automated Final Admission System is to enact document redesign and an automated management program. The proposed "JUST undergraduate final admission system" would store, route, and retrieve prospective and current application documents. The development of science and technology has immensely contributed to the growth of computers and the internet which has adversely increased the use and need for an automated applicant's admission system.

Universities, Colleges, Companies, organizations, etc. are seeking better ways to distribute and disseminate information throughout the world. In order to achieve this, they develop and rely on online applications, which help them input, update, process, and disseminate information. The internet has increased the use of online applications because data could be easily and readily accessed as well as documented. This project is targeted at allowing admission of applicants' decisions to be made faster and more efficiently. It looks at improving the present traditional system of the manual process (paper-based) which is manually documented. This type of project is a hot topic in the whole world to digitalize our education system.

1.2 Motivation

1.2.1 Existing System

There are several automated systems used for managing final admission processes in educational institutions. These systems differ in features, capabilities, and implementation depending on institutional needs and available resources.

Currently, JUST provides a web-based application titled "JUST Admission 2025". While this system includes some useful functionalities, it is not a fully integrated or complete solution. Several important features are missing, such as admission confirmation, sending admission confirmation messages, and verification of required documents.

Another issue in the current system is that any user can register regardless of whether they have been selected for admission or not. This creates inconsistency and reduces system reliability. Additionally, the system lacks proper validation mechanisms, which makes it difficult to ensure the authenticity and correctness of submitted data.

1.2.2 Limitations of the System

The current system suffers from several limitations that affect its efficiency and usability.

First, there is no proper validation process implemented. As a result, users can input incorrect or incomplete data without restriction. Second, the system allows unrestricted data insertion but does not provide proper mechanisms for updating or removing data, which can lead to data inconsistency.

Third, the system architecture and code structure are not well organized, making it difficult to maintain, scale, or extend the system in the future.

Another significant limitation is that users who are not eligible (i.e., those who did not get a chance for admission) can still register in the system. This reduces the accuracy and trustworthiness of the admission process.

Finally, key features such as admission confirmation, automated message sending, and document verification are either missing or not fully implemented, which prevents the system from functioning as a complete automated admission solution.

1.3 Objectives

The primary objective of this project is to develop a user-friendly and reliable web-based admission system that ensures maximum accuracy and efficiency. The system aims to provide the following facilities:

- **Efficiency:** Automate the admission process to reduce manual effort, save time, and streamline data handling.
- **Accuracy:** Minimize errors by reducing manual data entry and ensuring proper handling of application data and documents.
- **Consistency:** Ensure that all data and applications are processed uniformly, reducing the risk of inconsistency and data loss.
- **Validation and Reliability:** Implement proper validation mechanisms to ensure that only eligible candidates can register and submit accurate information.
- **Improved Performance:** Enhance system performance and reliability by using modern technologies and optimized system architecture.
- **Improved Security:** Strengthen system security by incorporating updated security measures to protect user data and prevent unauthorized access.
- **Cost Effectiveness:** Reduce operational costs by minimizing the need for manual labor and paper-based processes.
- **Objectivity:** Ensure a fair and unbiased admission process by evaluating applicants based on predefined and objective criteria.

Chapter 2

Software Requirement Specification

2.1 Introduction

2.1.1 Purpose

The primary purpose of the JUST Undergraduate Final Admission System (JUST-UFAS) is to provide a secure, efficient, and user-friendly platform for completing the final admission process. The previous manual and semi-automated system had several limitations, including lack of proper validation, poor code quality, absence of structured design patterns, and no mechanism to enforce admission deadlines.
This system is the first structured and fully integrated solution developed by the JUST Undergraduate Admission Committee. It aims to eliminate existing issues by introducing proper validation, deadline control, role-based access, and a more maintainable and scalable architecture.
This Software Requirement Specification (SRS) document describes the complete workflow of the final admission process, including:

- Applicant interaction through online application form submission.
- Verification and clearance by the Dean Office with required documents.
- Bank receipt verification by the Hall Office.
- Medical examination and reporting by the Medical Center.
- Final approval and document collection by the Registrar Office.

- Administrative control for managing users, roles, and system operations.

2.1.2 Document Conventions

The following conventions are used throughout this SRS document:

- Section headings are numbered hierarchically.
- **Bold text** indicates important terms or concepts.
- All figures and tables are labeled with appropriate names and numbering.
- The following abbreviations are used:
  - JUST: Jashore University of Science and Technology.
  - SRS: Software Requirements Specification.
  - JUST-UFAS: JUST Undergraduate Final Admission System.

2.1.3 Intended Audience and Reading Suggestions

This document follows IEEE SRS standards and is intended for:

- Admission Committee members
- Project Supervisor
- Chairman of the CSE Department
- Internal and External Viva Board Members
- Future Developers and System Maintainers
- Any stakeholders interested in understanding the system

Readers are encouraged to begin with the Introduction and Overall Description sections before proceeding to detailed requirements.

2.2 Overall Description

2.2.1 Product Perspective

The JUST-UFAS is a web-based automated admission management system designed to replace the existing manual and partially digital processes. The previous system lacked critical features such as proper validation mechanisms, deadline enforcement, and structured system architecture.

The proposed system introduces:

- Strong input validation to prevent invalid or incomplete submissions.
- Deadline control to restrict registration and updates after the admission deadline.
- Role-based access control to ensure secure and authorized operations.
- A modular and maintainable architecture following modern design patterns.
- Real-time tracking of application progress across different offices.
- Scalability to handle a large number of concurrent applicants.

2.2.2 Product Functions

The major functionalities of the JUST-UFAS are as follows:

- **Admin Module:**
  - Manage users and roles.
  - Control system operations (open/close admission).
  - Monitor overall admission progress.
- **Dean Office Module:**
  - Verify documents and validate applicants.
  - Approve or reject applications.
  - Export applicant data.
- **Hall Office Module:**
  - Verify bank receipts.

  - Manage residential clearance.
- **Medical Center Module:**
  - Record medical data.
  - Generate reports and provide clearance.
- **Registrar Office Module:**
  - Final admission approval.
  - Document collection and archiving.
- **Applicant Module:**
  - Registration and profile management.
  - Application form submission.
  - Document upload and download.

2.3 User Classes and Characteristics

The system includes multiple user roles, each with specific responsibilities. All users are expected to have basic knowledge of computers and web browsers.
Admin:

- Full system control.
- Manage users and roles.
- Control admission deadlines.

Dean Office:

- Verify documents.
- Approve or reject applications.

Hall Office:

- Verify payment and clearance.

Medical Office:

- Generate medical reports.

Registrar Office:

- Final admission approval.

Applicant:

- Submit application.
- Upload documents.
- Track status.

2.4 Constraints

The system operates under the following constraints:

- Admission is strictly controlled by deadlines.
- No registration or updates allowed after deadline.
- All inputs must pass validation rules.
- Role-based access control is enforced.
- Requires stable internet connection.
- Security measures must protect user data.

2.5 External Interface Requirements

2.5.1 User Interfaces

The system provides a web-based user interface designed for simplicity and usability. Interfaces are role-based, ensuring that each user (Admin, Dean Office, Hall Office, Medical Office, Registrar Office, Applicant) can only access permitted features.
The interface includes:

- Clear navigation and dashboard
- Form validation with proper instructions
- Role-specific panels for efficient task completion

2.5.2 Integration with Existing Data

The system supports integration with pre-existing admission data provided by the administration in Excel format. The admin uploads this data, which is then used to generate applicant accounts.

2.5.3 Integration with Internal Modules

All modules (Dean, Hall, Medical, Registrar) are interconnected and share a centralized database to ensure real-time updates and consistency.

2.5.4 Authentication and Authorization

The system implements secure authentication and role-based authorization:

- Admin provides login credentials to users
- Applicants log in using GST Application ID
- Access control ensures users can only perform authorized actions

2.6 System Features: Admin Management

2.6.1 Description and Priority

The Admin module has the highest priority and controls the entire system. The admin uploads applicant data (Excel file), manages users, controls admission deadlines, and monitors all activities.

2.6.2 Functional Requirements

Authentication

- Input: Username and password
- Output: Secure login access

Upload Applicant Data

- Input: Excel file containing applicant information
- Output: System generates applicant accounts

Manage Users

- View, update, and delete users

Update Applicant Information

- Modify applicant data if required

Control Admission

- Open or close admission system

View Reports

- Monitor admission progress and seat status

2.7 System Features: Dean Office

2.7.1 Description

The Dean Office verifies applicant information and approves or rejects applications based on eligibility criteria.

2.7.2 Functional Requirements

- Search applicant using ID
- View applicant profile
- Verify documents
- Approve or reject applicant
- View applicant lists (all, pending, approved)

2.8 System Features: Hall Office

2.8.1 Description

The Hall Office verifies payment and residential clearance.

2.8.2 Functional Requirements

- Search applicant
- View profile
- Verify bank receipt
- Approve applicant

2.9 System Features: Medical Office

2.9.1 Description

The Medical Office handles physical examination and medical clearance.

2.9.2 Functional Requirements

- Search applicant
- Insert medical data
- Generate report
- Approve applicant

2.10 System Features: Registrar Office

2.10.1 Description

The Registrar Office gives final admission approval and stores documents.

2.10.2 Functional Requirements

- Search applicant
- Receive documents
- Final approval
- Generate reports

2.11 System Features: Applicant

2.11.1 Description

Applicants do not register manually. Their accounts are created by admin-uploaded data. Applicants log in, complete their profile, and download the application form.

2.11.2 Functional Requirements

- Login using provided credentials
- Complete application form
- Upload required documents
- Download application form

2.12 Non-Functional Requirements

2.12.1 Performance

The system should handle a large number of users efficiently.

2.12.2 Security

Sensitive data must be protected using authentication and authorization.

2.12.3 Scalability

The system should support increasing users without performance loss.

2.12.4 Maintainability

The system should be easy to update and maintain.

2.12.5 Usability

The system must be user-friendly and easy to operate.

2.12.6 Reliability

The system should ensure high availability and minimal downtime.